

CHAPTER I

THE SCIENCE OF PSYCHOLOGY

Many people believe that women tend to talk more than men—with some even suggesting that this difference has a biological basis. One widely cited estimate is that women speak 20,000 words per day on average and men speak only 7,000. This claim seems plausible, but is it *true*? A group of psychologists led by Matthias Mehl decided to find out. They checked to see if anyone had actually tried to count the daily number of words spoken by women and men. No one had. So these researchers conducted a study in which female and male college students (369 in all) wore audio recorders while they went about their lives. The result? The women spoke an average of 16,215 words per day and the men spoke an average of 15,669—an extremely small difference that could easily be explained by chance. In an article in the journal *Science*, these researchers summed up their findings as follows: “We therefore conclude, on the basis of available empirical evidence, that the widespread and highly publicized stereotype about female talkativeness is unfounded” (Mehl, Vazire, Ramirez-Esparza, Slatcher, & Pennebaker, 2007, p. 82)¹.

Psychology is usually defined as the scientific study of human behavior and mental processes, and this example illustrates the features that make it scientific. In this chapter, we look closely at these features, review the goals of psychology, and address several basic questions that students often have about it. Who conducts scientific research in psychology? Why? Does scientific psychology tell us anything that common sense does not? Why should I bother to learn the scientific approach—especially if I want to be a clinical psychologist and not a researcher? These are extremely good questions, and answering them now will provide a solid foundation for learning the rest of the material in your course.

Notes

1. Mehl, M. R., Vazire, S., Ramirez-Esparza, N., Slatcher, R. B., & Pennebaker, J. W. (2007). Are women really more talkative than men? *Science*, 317, 82.

I. Methods of Knowing

Learning Objectives

1. Describe the 5 methods of acquiring knowledge
2. Understand the benefits and problems with each.

Take a minute to ponder some of what you know and how you acquired that knowledge. Perhaps you know that you should make your bed in the morning because your mother or father told you this is what you should do, perhaps you know that swans are white because all of the swans you have seen are white, or perhaps you know that your friend is lying to you because she is acting strange and won't look you in the eye. But should we trust knowledge from these sources? The methods of acquiring knowledge can be broken down into five categories each with its own strengths and weaknesses.

Intuition

The first method of knowing is intuition. When we use our intuition, we are relying on our guts, our emotions, and/or our instincts to guide us. Rather than examining facts or using rational thought, intuition involves believing what feels true. The problem with relying on intuition is that our intuitions can be wrong because they are driven by cognitive and motivational biases rather than logical reasoning or scientific evidence. While the strange behavior of your friend may lead you to think s/he is lying to you it may just be that s/he is holding in a bit of gas or is preoccupied with some other issue that is irrelevant to you. However, weighing alternatives and thinking of all the different possibilities can be paralyzing for some people and sometimes decisions based on intuition are actually superior to those based on analysis (people interested in this idea should read Malcolm Gladwell's book *Blink*)¹.

Authority

Perhaps one of the most common methods of acquiring knowledge is through authority. This method involves accepting new ideas because some authority figure states that they are true. These authorities include parents, the media, doctors, Priests and other religious authorities, the government, and professors. While in an ideal world we should be able to trust authority figures, history has taught us otherwise and many instances of atrocities against humanity are a consequence of people unquestioningly following

authority (e.g., Salem Witch Trials, Nazi War Crimes). On a more benign level, while your parents may have told you that you should make your bed in the morning, making your bed provides the warm damp environment in which mites thrive. Keeping the sheets open provides a less hospitable environment for mites. These examples illustrate that the problem with using authority to obtain knowledge is that they may be wrong, they may just be using their intuition to arrive at their conclusions, and they may have their own reasons to mislead you. Nevertheless, much of the information we acquire is through authority because we don't have time to question and independently research every piece of knowledge we learn through authority. But we can learn to evaluate the credentials of authority figures, to evaluate the methods they used to arrive at their conclusions, and evaluate whether they have any reasons to mislead us.

Rationalism

Rationalism involves using logic and reasoning to acquire new knowledge. Using this method premises are stated and logical rules are followed to arrive at sound conclusions. For instance, if I am given the premise that all swans are white and the premise that this is a swan then I can come to the rational conclusion that this swan is white without actually seeing the swan. The problem with this method is that if the premises are wrong or there is an error in logic then the conclusion will not be valid. For instance, the premise that all swans are white is incorrect; there are black swans in Australia. Also, unless formally trained in the rules of logic it is easy to make an error. Nevertheless, if the premises are correct and logical rules are followed appropriately then this is sound means of acquiring knowledge.

Empiricism

Empiricism involves acquiring knowledge through observation and experience. Once again many of you may have believed that all swans are white because you have only ever seen white swans. For centuries people believed the world is flat because it appears to be flat. These examples and the many visual illusions that trick our senses illustrate the problems with relying on empiricism alone to derive knowledge. We are limited in what we can experience and observe and our senses can deceive us. Moreover, our prior experiences can alter the way we perceive events. Nevertheless, empiricism is at the heart of the scientific method. Science relies on observations. But not just any observations, science relies on structured observations which is known as systematic empiricism.

The Scientific Method

The **scientific method** is a process of systematically collecting and evaluating evidence to test ideas and answer questions. While scientists may use intuition, authority, rationalism, and empiricism to generate new ideas they don't stop there. Scientists go a step further by using systematic empiricism to make careful

observations under various controlled conditions in order to test their ideas and they use rationalism to arrive at valid conclusions. While the scientific method is the most likely of all of the methods to produce valid knowledge, like all methods of acquiring knowledge it also has its drawbacks. One major problem is that it is not always feasible to use the scientific method; this method can require considerable time and resources. Another problem with the scientific method is that it cannot be used to answer all questions. As described in the following section, the scientific method can only be used to address empirical questions. This book and your research methods course are designed to provide you with an in-depth examination of how psychologists use the scientific method to advance our understanding of human behavior and the mind.

Notes

1. Gladwell, M. E. (2005). *Blink: The power of thinking without thinking*. (9th ed.). New York: Little, Brown & Co.

2. Understanding Science

Learning Objectives

1. Define science.
2. Describe the three fundamental features of science.
3. Explain why psychology is a science.
4. Define pseudoscience and give some examples.

What Is Science?

Some people are surprised to learn that psychology is a **science**. They generally agree that astronomy, biology, and chemistry are sciences but wonder what psychology has in common with these other fields. Before answering this question, however, it is worth reflecting on what astronomy, biology, and chemistry have in common with *each other*. It is clearly not their subject matter. Astronomers study celestial bodies, biologists study living organisms, and chemists study matter and its properties. It is also not the equipment and techniques that they use. Few biologists would know what to do with a radio telescope, for example, and few chemists would know how to track a moose population in the wild. For these and other reasons, philosophers and scientists who have thought deeply about this question have concluded that what the sciences have in common is a general approach to understanding the natural world. Psychology is a science because it takes this same general approach to understanding one aspect of the natural world: human behavior.

Features of Science

The general scientific approach has three fundamental features (Stanovich, 2010)¹. The first is **systematic empiricism**. Empiricism refers to learning based on observation, and scientists learn about the natural world systematically, by carefully planning, making, recording, and analyzing observations of it. As we will see, logical reasoning and even creativity play important roles in science too, but scientists are unique in their insistence on checking their ideas about the way the world is against their systematic observations. Notice, for example, that Mehl and his colleagues did not trust other people's stereotypes or even their own informal observations. Instead, they systematically recorded, counted, and compared the number of words spoken by a large sample of women and men. Furthermore, when their systematic observations turned out to conflict with people's stereotypes, they trusted their systematic observations.

The second feature of the scientific approach—which follows in a straightforward way from the first—is that it is concerned with **empirical questions**. These are questions about the way the world actually is and, therefore, can be answered by systematically observing it. The question of whether women talk more than men is empirical in this way. Either women really do talk more than men or they do not, and this can be determined by systematically observing how much women and men actually talk. Having said this, there are many interesting and important questions that are not empirically testable and that science is not in a position to answer. Among these are questions about values—whether things are good or bad, just or unjust, or beautiful or ugly, and how the world *ought* to be. So although the question of whether a stereotype is accurate or inaccurate is an empirically testable one that science can answer, the question—or, rather, the value judgment—of whether it is wrong for people to hold inaccurate stereotypes is not. Similarly, the question of whether criminal behavior has a genetic basis is an empirical question, but the question of what actions ought to be considered illegal is not. It is especially important for researchers in psychology to be mindful of this distinction.

The third feature of science is that it creates **public knowledge**. After asking their empirical questions, making their systematic observations, and drawing their conclusions, scientists publish their work. This usually means writing an article for publication in a professional journal, in which they put their research question in the context of previous research, describe in detail the methods they used to answer their question, and clearly present their results and conclusions. Increasingly, scientists are opting to publish their work in open access journals, in which the articles are freely available to all – scientists and nonscientists alike. This important choice allows publicly-funded research to create knowledge that is truly public.

Publication is an essential feature of science for two reasons. One is that science is a social process—a large-scale collaboration among many researchers distributed across both time and space. Our current scientific knowledge of most topics is based on many different studies conducted by many different researchers who have shared their work publicly over many years. The second is that publication allows science to be self-correcting. Individual scientists understand that, despite their best efforts, their methods can be flawed and their conclusions incorrect. Publication allows others in the scientific community to detect and correct these errors so that, over time, scientific knowledge increasingly reflects the way the world actually is.

A good example of the self-correcting nature of science is the “[Many Labs Replication Project](#)” – a large and coordinated effort by prominent psychological scientists around the world to attempt to replicate findings from 13 classic and contemporary studies (Klein et al., 2013)². One of the findings selected by these researchers for replication was the fascinating effect, first reported by Simone Schnall and her colleagues at the University of Plymouth, that washing one’s hands leads people to view moral transgressions—ranging from keeping money inside a found wallet to using a kitten for sexual arousal—as less wrong (Schnall, Benton, & Harvey, 2008)³. If reliable, this effect might help explain why so many religious traditions associate physical cleanliness with moral purity. However, despite using the same materials and nearly identical procedures with a much larger sample, the “Many Labs” researchers were unable to replicate the original finding (Johnson, Cheung, & Donnellan, 2013)⁴, suggesting that the original finding may have stemmed from the relatively small sample size (which can lead to unreliable results) used in the original study. To be clear, at this stage we are still unable to definitively conclude that the handwashing effect does not exist; however,

the effort that has gone into testing its reliability certainly demonstrates the collaborative and cautious nature of scientific progress.

For more on the replication crisis in psychology see: <http://nobaproject.com/modules/the-replication-crisis-in-psychology>

Science Versus Pseudoscience

Pseudoscience refers to activities and beliefs that are claimed to be scientific by their proponents—and may appear to be scientific at first glance—but are not. Consider the theory of biorhythms (not to be confused with sleep cycles or circadian rhythms that do have a scientific basis). The idea is that people's physical, intellectual, and emotional abilities run in cycles that begin when they are born and continue until they die. Allegedly, the physical cycle has a period of 23 days, the intellectual cycle a period of 33 days, and the emotional cycle a period of 28 days. So, for example, if you had the option of when to schedule an exam, you would want to schedule it for a time when your intellectual cycle will be at a high point. The theory of biorhythms has been around for more than 100 years, and you can find numerous popular books and websites about biorhythms, often containing impressive and scientific-sounding terms like *sinusoidal wave* and *bioelectricity*. The problem with biorhythms, however, is that scientific evidence indicates they do not exist (Hines, 1998)⁵.

A set of beliefs or activities can be said to be pseudoscientific if (a) its adherents claim or imply that it is scientific but (b) it lacks one or more of the three features of science. For instance, it might lack systematic empiricism. Either there is no relevant scientific research or, as in the case of biorhythms, there is relevant scientific research but it is ignored. It might also lack public knowledge. People who promote the beliefs or activities might claim to have conducted scientific research but never publish that research in a way that allows others to evaluate it.

A set of beliefs and activities might also be pseudoscientific because it does not address empirical questions. The philosopher Karl Popper was especially concerned with this idea (Popper, 2002)⁶. He argued more specifically that any scientific claim must be expressed in such a way that there are observations that would—if they were made—count as evidence against the claim. In other words, scientific claims must be **falsifiable**. The claim that women talk more than men is falsifiable because systematic observations could reveal either that they do talk more than men or that they do not. As an example of an unfalsifiable claim, consider that many people who believe in extrasensory perception (ESP) and other psychic powers claim that such powers can disappear when they are observed too closely. This makes it so that no possible observation would count as evidence against ESP. If a careful test of a self-proclaimed psychic showed that she predicted the future at better-than-chance levels, this would be consistent with the claim that she had psychic powers. But if she failed to predict the future at better-than-chance levels, this would also be consistent with the claim because her powers can supposedly disappear when they are observed too closely.

Why should we concern ourselves with pseudoscience? There are at least three reasons. One is that learning about pseudoscience helps bring the fundamental features of science—and their importance—into sharper focus. A second is that biorhythms, psychic powers, astrology, and many other pseudoscientific beliefs

are widely held and are promoted on the Internet, on television, and in books and magazines. Far from being harmless, the promotion of these beliefs often results in great personal toll as, for example, believers in pseudoscience opt for “treatments” such as homeopathy for serious medical conditions instead of empirically-supported treatments. Learning what makes them pseudoscientific can help us to identify and evaluate such beliefs and practices when we encounter them. A third reason is that many pseudosciences purport to explain some aspect of human behavior and mental processes, including biorhythms, astrology, graphology (handwriting analysis), and magnet therapy for pain control. It is important for students of psychology to distinguish their own field clearly from this “pseudo psychology.”

The Skeptic's Dictionary

An excellent source for information on pseudoscience is *The Skeptic's Dictionary* (<http://www.skeptdic.com>). Among the pseudoscientific beliefs and practices you can learn about are the following:

- **Cryptozoology.** The study of “hidden” creatures like Bigfoot, the Loch Ness monster, and the chupacabra.
- **Pseudoscientific psychotherapies.** Past-life regression, rebirthing therapy, and bioscream therapy, among others.
- **Homeopathy.** The treatment of medical conditions using natural substances that have been diluted sometimes to the point of no longer being present.
- **Pyramidology.** Odd theories about the origin and function of the Egyptian pyramids (e.g., that they were built by extraterrestrials) and the idea that pyramids, in general, have healing and other special powers.

Another excellent online resource is *Neurobonkers* (<http://neurobonkers.com>), which regularly posts articles that investigate claims that pertain specifically to psychological science.

Notes

1. Stanovich, K. E. (2010). *How to think straight about psychology* (9th ed.). Boston, MA: Allyn & Bacon.
2. Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Bahník, S., Bernstein, M. J., . . . Nosek, B. A. (2013). Investigating variation in replicability: A “many labs” replication project. *Social Psychology*, 45(3), 142-152. doi: 10.1027/1864-9335/a000178
3. Schnall, S., Benton, J., & Harvey, S. (2008). With a clean conscience: Cleanliness reduces the severity of moral judgments. *Psychological Science*, 19(12), 1219-1222. doi: 10.1111/j.1467-9280.2008.02227.x
4. Johnson, D. J., Cheung, F., & Donnellan, M. B. (2013). Does cleanliness influence moral judgments? A direct replication of Schnall, Benton, and Harvey (2008). *Social Psychology*, 45(3), 209-215. doi: 10.1027/1864-9335/a000186
5. Hines, T. M. (1998). Comprehensive review of biorhythm theory. *Psychological Reports*, 83, 19-64.
6. Popper, K. R. (2002). *Conjectures and refutations: The growth of scientific knowledge*. New York, NY: Routledge.

3. Goals of Science

Learning Objectives

1. Describe the three goals of science and give an example for each.
2. Distinguish between basic research and applied research.

The Broader Purposes of Scientific Research in Psychology

People have always been curious about the natural world, including themselves and their behavior (in fact, this is probably why you are studying psychology in the first place). Science grew out of this natural curiosity and has become the best way to achieve detailed and accurate knowledge. Keep in mind that most of the phenomena and theories that fill psychology textbooks are the products of scientific research. In a typical introductory psychology textbook, for example, one can learn about specific cortical areas for language and perception, principles of classical and operant conditioning, biases in reasoning and judgment, and people's surprising tendency to obey those in positions of authority. And scientific research continues because what we know right now only scratches the surface of what we *can* know.

The Three Goals of Science

The first and most basic goal of science is **to describe**. This goal is achieved by making careful observations. As an example, perhaps I am interested in better understanding the medical conditions that medical marijuana patients use marijuana to treat. In this case, I could try to access records at several large medical marijuana licensing centers to see which conditions people are getting licensed to use medical marijuana. Or I could survey a large sample of medical marijuana patients and ask them to report which medical conditions they use marijuana to treat or manage. Indeed, research involving surveys of medical marijuana patients has been conducted and has found that the primary symptom medical marijuana patients use marijuana to treat is pain, followed by anxiety and depression (Sexton, Cuttler, Finnell, & Mischley, 2016).¹

The second goal of science is **to predict**. Once we have observed with some regularity that two behaviors or events are systematically related to one another we can use that information to predict whether an event or behavior will occur in a certain situation. Once I know that most medical marijuana patients use marijuana to treat pain I can use that information to predict that an individual who uses medical marijuana likely experiences pain. Of course, my predictions will not be 100% accurate but if the relationship between medical marijuana use and pain is strong then my predictions will have greater than chance accuracy.

The third and ultimate goal of science is **to explain**. This goal involves determining the causes of behavior. For example, researchers might try to understand the mechanisms through which marijuana reduces pain. Does marijuana reduce inflammation which in turn reduces pain? Or does marijuana simply reduce the distress associated with pain rather than reducing pain itself? As you can see these questions tap at the underlying mechanisms and causal relationships.

Basic versus Applied Research

Scientific research is often classified as being either basic or applied. **Basic research** in psychology is conducted primarily for the sake of achieving a more detailed and accurate understanding of human behavior, without necessarily trying to address any particular practical problem. The research of Mehl and his colleagues falls into this category. **Applied research** is conducted primarily to address some practical problem. Research on the effects of cell phone use on driving, for example, was prompted by safety concerns and has led to the enactment of laws to limit this practice. Although the distinction between basic and applied research is convenient, it is not always clear-cut. For example, basic research on sex differences in talkativeness could eventually have an effect on how marriage therapy is practiced, and applied research on the effect of cell phone use on driving could produce new insights into basic processes of perception, attention, and action.

Notes

1. Sexton, M., Cuttler, C., Finnell, J., & Mischley, L (2016). A cross-sectional survey of medical cannabis users: Patterns of use and perceived efficacy. *Cannabis and Cannabinoid Research*, 1, 131-138. doi: 10.1089/can.2016.0007.

4. Science and Common Sense

Learning Objectives

1. Explain the limitations of common sense when it comes to achieving a detailed and accurate understanding of human behavior.
2. Give several examples of common sense or folk psychology that are incorrect.
3. Define skepticism and its role in scientific psychology.

Can We Rely on Common Sense?

Some people wonder whether the scientific approach to psychology is necessary. Can we not reach the same conclusions based on common sense or intuition? Certainly we all have intuitive beliefs about people's behavior, thoughts, and feelings—and these beliefs are collectively referred to as **folk psychology**. Although much of our folk psychology is probably reasonably accurate, it is clear that much of it is not. For example, most people believe that anger can be relieved by “letting it out”—perhaps by punching something or screaming loudly. Scientific research, however, has shown that this approach tends to leave people feeling more angry, not less (Bushman, 2002)¹. Likewise, most people believe that no one would confess to a crime that they had not committed unless perhaps that person was being physically tortured. But again, extensive empirical research has shown that false confessions are surprisingly common and occur for a variety of reasons (Kassin & Gudjonsson, 2004)².

Some Great Myths

In *50 Great Myths of Popular Psychology*, psychologist Scott Lilienfeld and colleagues discuss several widely held commonsense beliefs about human behavior that scientific research has shown to be incorrect (Lilienfeld, Lynn, Ruscio, & Beyerstein, 2010)³. Here is a short list:

- “People use only 10% of their brain power.”
- “Most people experience a midlife crisis in their 40's or 50's.”
- “Students learn best when teaching styles are matched to their learning styles.”
- “Low self-esteem is a major cause of psychological problems.”

- “Psychiatric admissions and crimes increase during full moons.”

How Could We Be So Wrong?

How can so many of our intuitive beliefs about human behavior be so wrong? Notice that this is an empirical question, and it just so happens that psychologists have conducted scientific research on it and identified many contributing factors (Gilovich, 1991)⁴. One is that forming detailed and accurate beliefs requires powers of observation, memory, and analysis to an extent that we do not naturally possess. It would be nearly impossible to count the number of words spoken by the women and men we happen to encounter, estimate the number of words they spoke per day, average these numbers for both groups, and compare them—all in our heads. This is why we tend to rely on mental shortcuts (what psychologists refer to as **heuristics**) in forming and maintaining our beliefs. For example, if a belief is widely shared—especially if it is endorsed by “experts”—and it makes intuitive sense, we tend to assume it is true. This is compounded by the fact that we then tend to focus on cases that confirm our intuitive beliefs and not on cases that dis-confirm them. This is called **confirmation bias**. For example, once we begin to believe that women are more talkative than men, we tend to notice and remember talkative women and silent men but ignore or forget silent women and talkative men. We also hold incorrect beliefs in part because it would be nice if they *were* true. For example, many people believe that calorie-reducing diets are an effective long-term treatment for obesity, yet a thorough review of the scientific evidence has shown that they are not (Mann et al., 2007)⁵. People may continue to believe in the effectiveness of dieting in part because it gives them hope for losing weight if they are obese or makes them feel good about their own “self-control” if they are not.

Scientists—especially psychologists—understand that they are just as susceptible as anyone else to intuitive but incorrect beliefs. This is why they cultivate an attitude of **skepticism**. Being skeptical does not mean being cynical or distrustful, nor does it mean questioning every belief or claim one comes across (which would be impossible anyway). Instead, it means pausing to consider alternatives and to search for evidence—especially systematically collected empirical evidence—when there is enough at stake to justify doing so. For example, imagine that you read a magazine article that claims that giving children a weekly allowance is a good way to help them develop financial responsibility. This is an interesting and potentially important claim (especially if you have children of your own). Taking an attitude of skepticism, however, would mean pausing to ask whether it might be instead that receiving an allowance merely teaches children to spend money—perhaps even to be more materialistic. Taking an attitude of skepticism would also mean asking what evidence supports the original claim. Is the author a scientific researcher? Is any scientific evidence cited? If the issue was important enough, it might also mean turning to the research literature to see if anyone else had studied it.

Because there is often not enough evidence to fully evaluate a belief or claim, scientists also cultivate a **tolerance for uncertainty**. They accept that there are many things that they simply do not know. For example, it turns out that there is no scientific evidence that receiving an allowance causes children to be

more financially responsible, nor is there any scientific evidence that it causes them to be materialistic. Although this kind of uncertainty can be problematic from a practical perspective—for example, making it difficult to decide what to do when our children ask for an allowance—it is exciting from a scientific perspective. If we do not know the answer to an interesting and empirically testable question, science, and perhaps even you as a researcher, may be able to provide the answer.

Notes

1. Bushman, B. J. (2002). Does venting anger feed or extinguish the flame? Catharsis, rumination, distraction, anger, and aggressive responding. *Personality and Social Psychology Bulletin*, 28, 724–731.
2. Kassin, S. M., & Gudjonsson, G. H. (2004). The psychology of confession evidence: A review of the literature and issues. *Psychological Science in the Public Interest*, 5, 33–67.
3. Lilienfeld, S. O., Lynn, S. J., Ruscio, J., & Beyerstein, B. L. (2010). *50 great myths of popular psychology*. Malden, MA: Wiley-Blackwell.
4. Gilovich, T. (1991). *How we know what isn't so: The fallibility of human reason in everyday life*. New York, NY: Free Press.
5. Mann, T., Tomiyama, A. J., Westling, E., Lew, A., Samuels, B., & Chatman, J. (2007). Medicare's search for effective obesity treatments: Diets are not the answer. *American Psychologist*, 62, 220–233.

5. Experimental and Clinical Psychologists

Learning Objectives

1. Define the clinical practice of psychology and distinguish it from experimental psychology.
2. Explain how science is relevant to clinical practice.
3. Define the concept of an empirically supported treatment and give some examples.

Who Conducts Scientific Research in Psychology?

Experimental Psychologists

Scientific research in psychology is generally conducted by people with doctoral degrees (usually the **doctor of philosophy [Ph.D.]**) and master's degrees in psychology and related fields, often supported by research assistants with bachelor's degrees or other relevant training. Some of them work for government agencies (e.g., doing research on the impact of public policies), national associations (e.g., the American Psychological Association), non-profit organizations (e.g., National Alliance on Mental Illness), or in the private sector (e.g., in product marketing and development; organizational behavior). However, the majority of them are college and university faculty, who often collaborate with their graduate and undergraduate students. Although some researchers are trained and licensed as clinicians for mental health work—especially those who conduct research in clinical psychology—the majority are not. Instead, they have expertise in one or more of the many other subfields of psychology: behavioral neuroscience, cognitive psychology, developmental psychology, personality psychology, social psychology, and so on. Doctoral-level researchers might be employed to conduct research full-time or, like many college and university faculty members, to conduct research in addition to teaching classes and serving their institution and community in other ways.

Of course, people also conduct research in psychology because they enjoy the intellectual and technical challenges involved and the satisfaction of contributing to scientific knowledge of human behavior. You might find that you enjoy the process too. If so, your college or university might offer opportunities to get involved in ongoing research as either a research assistant or a participant. Of course, you might find that you do not enjoy the process of conducting scientific research in psychology. But at least you will have a better understanding of where scientific knowledge in psychology comes from, an appreciation of its strengths and limitations, and an awareness of how it can be applied to solve practical problems in psychology and everyday life.

Scientific Psychology Blogs

A fun and easy way to follow current scientific research in psychology is to read any of the many excellent blogs devoted to summarizing and commenting on new findings. Among them are the following:

Research Digest, <http://digest.bps.org.uk/>

Talk Psych, <http://www.talkpsych.com/>

Brain Blogger, <http://brainblogger.com/>

Mind Hacks, <http://mindhacks.com/>

PsyBlog, <http://www.spring.org.uk>

You can also browse to <http://www.researchblogging.org>, select psychology as your topic, and read entries from a wide variety of blogs.

Clinical Psychologists

Psychology is the scientific study of behavior and mental processes. But it is also the application of scientific research to “help people, organizations, and communities function better” (American Psychological Association, 2011)¹. By far the most common and widely known application is the **clinical practice of psychology**—the diagnosis and treatment of psychological disorders and related problems. Let us use the term *clinical practice* broadly to refer to the activities of clinical and counseling psychologists, school psychologists, marriage and family therapists, licensed clinical social workers, and others who work with people individually or in small groups to identify and help address their psychological problems. It is important to consider the relationship between scientific research and clinical practice because many students are especially interested in clinical practice, perhaps even as a career.

The main point is that psychological disorders and other behavioral problems are part of the natural world. This means that questions about their nature, causes, and consequences are empirically testable and therefore subject to scientific study. As with other questions about human behavior, we cannot rely on our intuition or common sense for detailed and accurate answers. Consider, for example, that dozens of popular books and thousands of websites claim that adult children of alcoholics have a distinct personality profile, including low self-esteem, feelings of powerlessness, and difficulties with intimacy. Although this sounds plausible, scientific research has demonstrated that adult children of alcoholics are no more likely to have these problems than anybody else (Lilienfeld et al., 2010)². Similarly, questions about whether a particular psychotherapy is effective are empirically testable questions that can be answered by scientific research. If a new psychotherapy is an effective treatment for depression, then systematic observation should reveal that depressed people who receive this psychotherapy improve more than a similar group of depressed people

who do not receive this psychotherapy (or who receive some alternative treatment). Treatments that have been shown to work in this way are called **empirically supported treatments**.

Empirically Supported Treatments

An empirically supported treatment is one that has been studied scientifically and shown to result in greater improvement than no treatment, a placebo, or some alternative treatment. These include many forms of psychotherapy, which can be as effective as standard drug therapies. Among the forms of psychotherapy with strong empirical support are the following:

- **Acceptance and commitment therapy (ACT)**. for depression, mixed anxiety disorders, psychosis, chronic pain, and obsessive-compulsive disorder.
- **Behavioral couples therapy**. For alcohol use disorders.
- **Cognitive behavioral therapy (CBT)**. For many disorders including eating disorders, depression, anxiety disorders, etc.
- **Exposure therapy**. For post-traumatic stress disorder and phobias.
- **Exposure therapy with response prevention**. For obsessive-compulsive disorder.
- **Family-based treatment**. For eating disorders.

For a more complete list, see the following website, which is maintained by Division 12 of the American Psychological Association, the Society for Clinical Psychology: <http://www.div12.org/psychological-treatments>

Many in the clinical psychology community have argued that their field has not paid enough attention to scientific research—for example, by failing to use empirically supported treatments—and have suggested a variety of changes in the way clinicians are trained and treatments are evaluated and put into practice. Others believe that these claims are exaggerated and the suggested changes are unnecessary (Norcross, Beutler, & Levant, 2005)³. On both sides of the debate, however, there is agreement that a scientific approach to clinical psychology is essential if the goal is to diagnose and treat psychological problems based on detailed and accurate knowledge about those problems and the most effective treatments for them. So not only is it important for scientific research in clinical psychology to continue, but it is also important for clinicians who never conduct a scientific study themselves to be scientifically literate so that they can read and evaluate new research and make treatment decisions based on the best available evidence.

Notes

1. American Psychological Association. (2011). *About APA*. Retrieved from <http://www.apa.org/about>

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6. Key Takeaways and Exercises

Key Takeaways

- Knowledge is acquired in many ways including intuition, authority, rationalism, empiricism, and the scientific method
- Science is a general way of understanding the natural world. Its three fundamental features are systematic empiricism, empirical questions, and public knowledge.
- Psychology is a science because it takes the scientific approach to understanding human behavior.
- Pseudoscience refers to beliefs and activities that are claimed to be scientific but lack one or more of the three features of science. It is important to distinguish the scientific approach to understanding human behavior from the many pseudoscientific approaches.
- Psychologists conduct research in order to describe basic phenomenon, to make predictions about future behaviors, and to explain the causes of behavior.
- Basic research is conducted to learn about human behavior for its own sake, and applied research is conducted to solve some practical problem. Both are valuable, and the distinction between the two is not always clear-cut.
- People's intuitions about human behavior, also known as folk psychology, often turn out to be wrong. This is one primary reason that psychology relies on science rather than common sense.
- Researchers in psychology cultivate certain critical-thinking attitudes. One is skepticism. They search for evidence and consider alternatives before accepting a claim about human behavior as true. Another is tolerance for uncertainty. They withhold judgment about whether a claim is true or not when there is insufficient evidence to decide.
- Scientific research in psychology is conducted mainly by people with doctoral degrees in psychology and related fields, most of whom are college and university faculty members. They do so for professional and for personal reasons, as well as to contribute to scientific knowledge about human behavior. Most psychologists are experimental psychologists and they conduct research.
- The clinical practice of psychology—the diagnosis and treatment of psychological problems—is one important application of the scientific discipline of psychology.
- Scientific research is relevant to clinical practice because it provides detailed and accurate knowledge about psychological problems and establishes whether treatments are effective.

Exercises

- Practice: Consider three things you know and determine how you acquired that knowledge (authority, intuition, rationalism, empiricism, the scientific method).
- Practice: Try to generate different research questions to describe, predict, and explain a phenomenon that interests you.

- Practice: Based on your own experience or on things you have already learned about psychology, list three basic research questions and three applied research questions of interest to you.
- Practice: List three empirical questions about human behavior. List three nonempirical questions about human behavior.
- Practice: For each of the following intuitive beliefs about human behavior, list three reasons that it might be true and three reasons that it might not be true:
 - You cannot truly love another person unless you love yourself.
 - People who receive “crisis counseling” immediately after experiencing a traumatic event are better able to cope with that trauma in the long term.
 - Studying is most effective when it is always done in the same location.
- Watch the following video, in which psychologist Scott Lilienfeld talks about confirmation bias, tunnel vision, and using evidence to evaluate the world around us:



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- Reading in print? Go to <https://youtu.be/gKVe0-pWjA0> or scan this QR code with your phone:



- Discussion: Consider the following psychological claim. “People’s choice of spouse is strongly influenced by their perception of their own parents. Some choose a spouse who is similar in some way to one of their parents. Others choose a spouse who is different from one of their parents.” Is this claim falsifiable? Why or why not?
- Discussion: People sometimes suggest that psychology cannot be a science because either (a) human behavior cannot be predicted with perfect accuracy or (b) much of its subject matter (e.g., thoughts and feelings) cannot be observed directly. Do you agree or disagree with each of these ideas? Why?
- Watch the following video by PHD Comics for an overview of open access publishing and why it matters:



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- Discussion: Some clinicians argue that what they do is an “art form” based on intuition and personal experience and therefore cannot be evaluated scientifically. Write a paragraph about how satisfied you would be with such a clinician and why from each of three perspectives:
 - a potential client of the clinician
 - a judge who must decide whether to allow the clinician to testify as an expert witness in a child abuse case
 - an insurance company representative who must decide whether to reimburse the clinician for their services
- Practice: Create a short list of questions that a client could ask a clinician to determine whether they pay sufficient attention to scientific research.

Video Attributions

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